Transmission Network Planning Under Security and Environmental Constraints

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***Abstract*—This paper explores the impact of CO_2 emission trading on capacity planning of electric power transmission systems. Two different models for annual emission costs are assumed. The CO_2 emission price is modeled as a probability density function in the transmission network planning problem. The Monte Carlo technique is deployed to simulate the CO_2 emission price volatility. The transmission network planning problem is formulated as a mixed-integer optimization whose objective is to minimize the sum of annual generator operating costs and annuitized transmission investment costs over different demand levels subject to N-1 network security constraints as well as operating limits on system components. The overall problem is formulated within the framework of a linear dc optimal power flow incorporating binary decision variables to model the lumpy nature of transmission investment. A linear model of losses is also proposed and included in the dc power flow model. The proposed approach can be used to determine the most probable optimal transmission capacity. The methodology is demonstrated through case studies simulated on the IEEE 24-bus network.**

***Index Terms*—CO_2 emissions, emission trading market, mixed-integer optimization, Monte Carlo, transmission expansion, transmission losses.**

I. Introduction

After the emergence of the electricity market, especially in developed countries, there has been a transition from a vertically integrated and regulated power system structure to an unbundled, liberalized electricity industry. The new structure is intended to foster competition between generators and suppliers so as to create a more efficient electrical energy supply system. In this new structure, generation companies (GenCos) are in charge of commissioning and decommissioning power plants and hence are effectively responsible for generation planning. Transmission planning is conducted separately by transmission companies (TransCos). The coordination between Transmission and Generation planning which is an important benefit of the vertical integration is no longer applicable. The main consequence of new unbundled structure is an increase in uncertainties and the risk level of expansion proposals prepared by both GenCos and TransCos [1], [2]. In addition to these uncertainties posed by the electricity market, environmental constraints manifest in the new emissions market which was introduced in 2003 have made the planning procedure even more complicated. In October 2003, the European parliament approved a new emission trading scheme in order to meet the commitments made in the framework of the Kyoto Protocol. This trading scheme has been in operation since January 2005. Under this scheme there is a specific allowance of greenhouse gas (GHG) emissions for each European Union member and any excess emissions should be bought on the allowance trading market. The introduction of the emission market inevitably increased the total operation cost of power plants which consequently has influence on transmission network planning (TNP). Since this market has been in operation there have been considerable fluctuations in the CO_2 emission price [3] which signal to transmission network planners that they need to deal with another uncertain parameter, the CO_2 emission price. As the environmental constraint has only recently been introduced, the impact of CO_2 price on TNP has rarely been studied. Transmission networks, however, should be designed economically so that even with changes in CO_2 emission price they provide a nondiscriminatory environment to reduce the out-of-merit generations. Most studies are about the role of environmental constraints in either generation expansion [4] or generator maintenance scheduling [5]. One of the main aims of this paper, unlike previous works [6], [7], is to introduce the environmental constraint as a volatile operation cost to the objective function of TNP so that the most likely optimal transmission capacity is proposed.

An optimum proposal for transmission planning generally yields 1) a reduction in out-of-merit generator cost and 2) a robust transmission network against outages. It is worth noting that typically a considerable percentage of the capacities of transmission lines are usually held in reserve to satisfy network related contingencies. In [8] the TNP problem is solved without considering security constraints. However due to the profound effect of security constraints on TNP, in this paper the "N-1" security criterion has been taken into account.

Another important point that should be stressed is that the maximum required capacity of a transmission line does not necessarily occur under maximum load conditions. The generation pattern is changed optimally at different load levels. For example, at the low load level, low cost power plants operate, but the expensive units are not in service. This variation in generation dispatch is likely to cause some lines to experience higher power flow than they do at high load levels. Therefore, in this study two load levels are considered. Furthermore, in practice, there are specific transmission capacities which can be assumed for overhead lines, so the credible capacity of a transmission line is an integer variable. In order to take a practical approach, the TNP problem is formulated as a mixed-integer optimization problem so that only particular capacities can be allocated to candidate transmission lines.

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Another crucial point which should be considered in TNP is losses. Losses can simply cause a shift in optimal generation dispatch and consequently a change in optimum transmission line capacities. As a dc model is assumed for the network, a linear loss model is introduced and validated in this paper.

This paper is organized as follows. In the next section an overview of the emission market and elements affecting the emission allowance price is presented. In Section III the TNP problem is formulated and models are proposed for losses and CO_2 emission price. Then, in Section IV, the numerical example for security constrained TNP including environmental constraints is presented and discussed. Finally, some concluding remarks are given in Section V.

II. Emission Market

The European Union established the first carbon dioxide emission trading market in the world in 2005 in order to meet the Kyoto Protocol commitments. Under this trading system, participating companies can buy or sell their emission allowances. Consequently emission targets of these companies can be achieved at least cost. The European Union Emission Trading System (EU ETS) can be divided into three distinctive periods. The first period, from 2005 to 2007 was a trial period—learning by doing—to understand what this scheme was supposed to achieve. This first period was an opportunity for all participants to glean necessary experience in order to provide and develop the infrastructures for a successful cap-and-trade system. During the second period running from 1st January 2008 to 2012 capped national emissions were allocated to each member at an average of around 6.5% below the 2005 level. Finally, in the third period from 2013 to 2020, the EU's target is to lower GHG emission level to at least 20% of the level in 1990 [9].

The main lesson from the first period was that the CO_2 allowance price is highly volatile. A rapid change in allowance price in April 2006 is evidence of this volatility; the emission price fell from $32€/tCO_2$ to $10€/tCO_2$. This happened due to over-allocation of national allowances in the first period in 2005. In April 2006 when the real figures for each installation were released, the participants realized that there was a safe margin of allowance and this yielded a drop in the CO_2 price [3].

Although there is currently more available information about CO_2 emitted by installations than there was before, the CO_2 price can still be affected by several factors. These factors might be due to the following possible future changes in EU ETS.

1) In the new trading system there is no longer a free national allocation, but instead an emission cap is considered for all states as a whole. Therefore each state should bid for the emission allowance at auction.
2) After 2013 the allowance cap will decrease annually by a factor of 1.74% in a linear manner. The starting point is the average amount of allowance in the second period [6].
3) The plan is to involve more installations in emission trading each year.
4) So far among GHGs only CO_2 has been considered in the emission market, but according to the EU decision in the third period other GHGs will be added to the emission trading market.

During these three periods, therefore, participants are facing a dynamic market and there is more likely to be again either a dramatic drop or a price increase. Under these circumstances, power plants having a considerable contribution to GHG emissions can be affected by this dynamic market. For example in U.K. power plants having a 33% contribution to CO_2 emission fall under this category [10]. Any uncertainty in the emission allowance price leads to uncertainty in operating costs which ultimately has an impact on required transmission network capacity. Therefore, the presence of CO_2 emission trading is likely to increase the level of uncertainty in TNP. In this paper the CO_2 emission cost is incorporated in the TNP objective function and the effect of its volatility on transmission expansion is investigated.

III. Problem Formulation

A dc-based power flow model is applied as this model has been successfully used to solve TNP problems [1], [2], [6], [11], [12]. The main aim of TNP is to propose the best scenario of transmission expansion which minimizes the sum of transmission investment costs and operating costs in a given network for a horizon year. Investment cost represents the total annuitized cost of the required transmission capacity, whereas the operation cost consists of generation cost and emission cost. Furthermore, the optimum solution should be feasible for both intact and contingent networks. In other words, Kirchhoff's first and second laws should be respected, as should operating limits on system components. In practice, due to the lumpy nature of transmission investment, only particular capacities can be assumed for candidate lines, so discrete variables are incorporated into the TNP problem turning the problem into a mixed-integer optimization problem.

The objective function can therefore be expressed as follows:

$$\underset{G_i^j, \overline{P}_{\max i}}{Minimize}: \sum_{j=1}^{Np} D_j . \sum_{i=1}^{Ng} C_i . G_i^j + \sum_{i=1}^{Ng} \pi_{co2} . E_i + \sum_{i=1}^{\overline{N_l}} T_{ci} . l_i . \overline{P}_{\max i}. \quad (1)$$

In (1) the first term represents generation cost, the second term is the emission cost and the last term represents the transmission investment cost. Annual emission (E_i) is modeled in two different ways which will be elaborated in Section III-C. Notations in (1) are defined as follows:

D_j	duration of the load level j (hour);
C_i	incremental cost of generator i (€/MWh);
G_i^j	power generated by generator i at load level j (MW);
π_{co2}	emission allowance price (€/tCo2);
E_i	annual emission by generator i (tCo2/year);
T_{ci}	annuitized investment cost of line i (€/km.MW.year);
l_i	length of transmission line i (km);

$\overline{P}_{\max i}$ capacity of candidate transmission line i (MW);

$\overline{N_l}$ number of candidate transmission lines;

N_p number of load levels;

N_g number of generators.

If a set S_i containing N_i different credible capacities for candidate line i is considered then the capacity of candidate line i is calculated by (3):

$$S_i = \left\{\overline{P^n_{i1}}, \overline{P^n_{i2}}, \ldots, \overline{P^n_{ik}}, \ldots\right\} \quad i = 1, \ldots, \overline{Nl} \tag{2}$$

$$\overline{P}_{\max i} = \sum_{k=1}^{N_i} x_{ik}.\overline{P^n_{ik}} \quad i = 1, \ldots, \overline{Nl}. \tag{3}$$

In (3) x_{ik} is a binary decision variable, and $\overline{P^n_{ik}}$ is kth credible capacity of the set S_i. As only one out of N_i members of S_i must be chosen the equality constraint (4) is added to the problem:

$$X_i = \sum_{k=1}^{N_i} x_{ik} \leq 1 \quad i = 1, \ldots, \overline{Nl}. \tag{4}$$

It should be noted that the resultant X_i can be either 0 or 1.

Also, at each load level j the objective function is subject to the following constraints:

$$\sum_{\forall\ i \in N_{gk}} G^j_i + \sum_{\forall\ i \in N_{lk}} P^j_i - L^j_k = 0 \quad k = 1, \ldots, N_b \tag{5}$$

$$\sum_{i=1}^{N_g} G^j_i - \sum_{i=1}^{N_b} L^j_i - P^j_{loss} = 0 \tag{6}$$

where

P^j_i power flowing in transmission line i at load level j (MW);

L^j_k load demand at bus k for load level j (MW);

N_{gk} set of generators connected to bus k;

N_{lk} set of lines connected to bus k;

N_b number of buses;

P^j_{loss} total losses at load level j.

Equality constraint (5) is Kirchhoff's Current Law (KCL) and constraint (6) is the power balance in the whole network.

Furthermore, Kirchhoff's Voltage Law (KVL) should be respected. Different approaches, however, are taken into account to formulate KVL equations. For existing lines, in the p.u. system, KVL is expressed by equality constraint (7), whereas for candidate lines a disjunctive form with two inequality constraints given by (8) and (9) is constructed [7]:

$$P^j_i - \frac{\left(\theta^j_k - \theta^j_l\right)}{X_i} = 0 \quad i = 1, \ldots, N_l \tag{7}$$

$$P^j_i - \frac{\left(\theta^j_k - \theta^j_l\right)}{X_i} \leq M_k.(1 - X_i) \quad i = 1, \ldots, \overline{N_l} \tag{8}$$

$$P^j_i - \frac{\left(\theta^j_k - \theta^j_l\right)}{X_i} \geq M_k.(X_i - 1) \quad i = 1, \ldots, \overline{N_l} \tag{9}$$

where

M_k very large number;

N_l number of existing transmission lines;

θ^j_k voltage angle of bus k (radian);

X_i reactance of transmission line i (p.u.).

Limits on power system components including generation upper and lower bounds as well as maximum line capacities are expressed by (10)–(12):

$$G_{\min i} \leq G^j_i \leq G_{\max i} \quad i = 1, \ldots, N_g \tag{10}$$

$$-P_{\max i} \leq P^j_i \leq P_{\max i} \quad i = 1, \ldots, N_l \tag{11}$$

$$-\overline{P}_{\max i} \leq P^j_i \leq \overline{P}_{\max i} \quad i = 1, \ldots, \overline{N_l}. \tag{12}$$

The above problem is solved at different demand levels to reflect the temporal variations in demand and generation. Equation (11) and (12) shows that the maximum capacities of transmission lines ($P_{\max i}$ and $\overline{P}_{\max i}$) should have the same value at all load levels.

Regarding the above formulations G^j_i, x_{ik}, P^j_i and θ^j_k are decision variables.

In the following sub-sections more details regarding losses modeling, network security constraints and annual CO_2 emission modeling are given.

A. Modeling Losses

Loss allocation methods are still under discussion. One of the methods is to allocate the losses in each line to an additional bus assumed in the middle of the line [13]. The main drawback of this method is that it enlarges the dimension of the system and hence the problem. Other conventional methods usually consider a slack bus to which all losses are allocated [14]. However, all generators are supposed to participate in supplying the losses. Loss allocation should reflect the fact that the generators which impose more losses on the network should generate less than others. For a fair allocation, it is proposed to allocate the losses in each line equally to both sending and receiving end buses of the line as shown in Fig. 1.

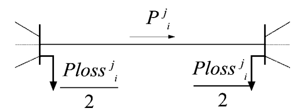

Fig. 1. Proposed model to allocate losses.

In Fig. 1 $Ploss^j_i$ is losses of line i at load level j.

Based on this allocation, the equality constraint (5) should be rewritten as shown in (13). Constraint (6) is no longer needed because (13) can also satisfy the load balance in the whole network:

$$\sum_{\forall\ i \in N_{gk}} G^j_i + \sum_{\forall\ i \in N_{lk}} P^j_i - L^j_k - \sum_{\forall\ i \in N_{lk}} \frac{Ploss^j_i}{2} = 0 \quad k = 1, \ldots, N_b. \tag{13}$$

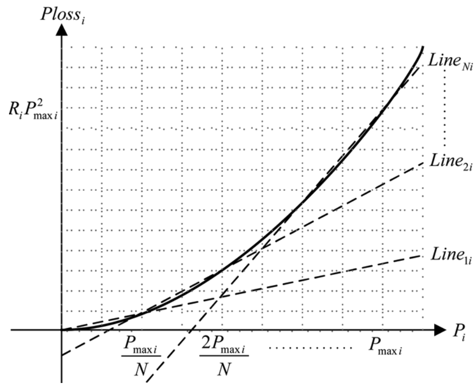

Fig. 2. Linear model of losses.

One of the methods widely used to calculate losses is the B-coefficients method. This method determines the total losses by a quadratic equation which is a function of operational system state, operating point of generators and network parameters [15]. However, to incorporate losses in a linear system defined by (13), B-coefficients cannot easily be employed. In [16] a piecewise linear model of the quadratic form of losses is presented, but the model is still complicated. Some binary variables are involved in the calculation requiring mixed-integer programming to solve the problem. A curve fitting technique is used in [17] to model the loss function with just a straight line. Obviously, this model is not accurate and cannot show the quadratic nature of losses properly.

In the following formulation a linear but accurate model for losses is introduced. Under the dc approximation, and in the p.u. system, the losses in a transmission line i whose resistance is R_i can be expressed by (14) [18]:

$$Ploss_i^j = R_i.(P_i^j)^2. \tag{14}$$

The above equation is a quadratic function, but can be expressed with a piecewise linear model. If a linear model consisting of N line pieces is assumed between $(0,0)$ and $(P_{\max i}, R_i P^2_{\max i})$ the equation of nth line piece $(Line^j_{ni})$ can be formulated by (15) (see Fig. 2):

$$Line^j_{ni} : \frac{R_i.P_{maxi}}{N} \times \left[(2n-1).P^j_{absi} + P_{\max i}.\frac{n-n^2}{N}\right], \; P^j_{absi} = \left|P^j_i\right|. \tag{15}$$

Then, the $Line^j_{ni}$ models $Ploss^j_i$ in the range shown by (16):

$$\frac{(n-1).P_{\max i}}{N} \leq P^j_i \leq \frac{n.P_{\max i}}{N} \quad n = 1,\ldots,N. \tag{16}$$

In order to select the proper line segment in this model based on (15), the following N inequality constraints for losses of line i at load level j are introduced to the main problem:

$$Ploss^j_i \geq Line^j_{ni} \quad n = 1,\ldots,N. \tag{17}$$

As all equations should be expressed in linear form, in order to produce absolute values of P^j_i used in (15), three inequalities given by (18) to (20) are employed:

$$P^j_{absi} \geq P^j_i \tag{18}$$
$$P^j_{absi} \geq -P^j_i \tag{19}$$
$$P^j_{absi} \geq 0. \tag{20}$$

It should be noted that the number of pieces (N) on the one hand boosts the accuracy and on the other hand may increase the dimension of the model. As shown in Section IV-B a three piecewise linear model is accurate enough for this study, although in the proposed method there is no limit to the number of pieces that can be used.

B. Security Constraint

Transmission network should be robust against any possible line outage. This robustness is usually guaranteed by introducing the so-called "N-1" criterion to the TNP problem [19]. For contingent networks, as for intact network, KCL and KVL are respected along with power flow limits. Due to generators' ramp rate limits, however, the power output of generators cannot be changed significantly once a line outage occurs. Therefore, in this study, generator outputs are assumed to be unchanged following a contingency in the network.

Also, for the sake of simplicity, a lossless model for contingent networks is assumed. It can be a sensible assumption as once a line trips, providing generators are not re-dispatched, losses have negligible effect on maximum required transmission capacities. Nevertheless, losses mostly do have an effect on generation pattern in normal condition.

In order to deduct losses from generators in an almost uniform manner for the contingent networks, for generator i a dummy load r^j_i is assumed at load level j with that the KCL at each bus can be written as (21). Equation (22) also secures the equality of summation of all dummy loads with total losses. See equations (21) and (22) at the bottom of the page, where c denotes a contingent network.

Equations (23) and (24) below are needed to ensure that just a small percentage (α) of each generator output is decreased and an inordinate amount of losses is not deducted from only

$$\sum_{\forall\, i \in N_{gk}} G^j_i + \sum_{\forall\, i \in N_{lk}, i \neq c} P^{j(c)}_i - L^j_k - \sum_{\forall\, i \in N_{gk}} r^j_i = 0 \quad k = 1,\ldots,N_b \tag{21}$$

$$\sum_{i=1}^{N_g} r^j_i - \sum_{i=1}^{Nl+\overline{Nl}} Ploss^j_i = 0 \tag{22}$$

one particular generator. In this study a value of 4% is assumed for α:

$$(G_i^j - r_i^j) \geq G_{\min i} \tag{23}$$
$$0 \leq r_i^j \leq \alpha . G_i^j . \tag{24}$$

Similar to the intact network, KVL for existing and candidate lines can mathematically be expressed by (25)–(27) at the bottom of the page.

C. CO_2 *Emission Modeling*

The CO_2 emission is modeled based on two different approaches. The first approach is to let all generators either sell their emission allowance surpluses or buy the shortage in allowance from the emission trading market. In this case, therefore, the operation cost of some generators may decrease as they benefit from selling the allowance surplus. This model is formulated as follows:

$$E_i = K_{co2i} . \sum_{j=1}^{Np} D_j . G_i^j - A_i \quad i = 1, \ldots, N_g \tag{28}$$

where

K_{CO2i} CO_2 emission factor of generator i (tCO_2/MWh);

A_i emission allowance of generator i ($tCO_2/year$).

The second approach is to ignore any benefit gained by power plants emitting less than their emission allowance. In this model, therefore, only power plants which exceed the emission allowance are penalized and should pay for the excessive emission based on the price in the emission trading market. Regarding this assumption the annual CO_2 emission of power plant is mathematically expressed as follows:

$$E_i \geq K_{co2i} . \sum_{j=1}^{Np} D_j . G_{ij} - A_i \quad i = 1, \ldots, N_g \tag{29}$$
$$E_i \geq 0. \tag{30}$$

These two models will be compared in Section V-B.

As described in Section II, due to upcoming policies and unpredictable drivers, CO_2 emission price may change unexpectedly in future leading to a variable generation operating cost which is an element of TNP's objective function (1).

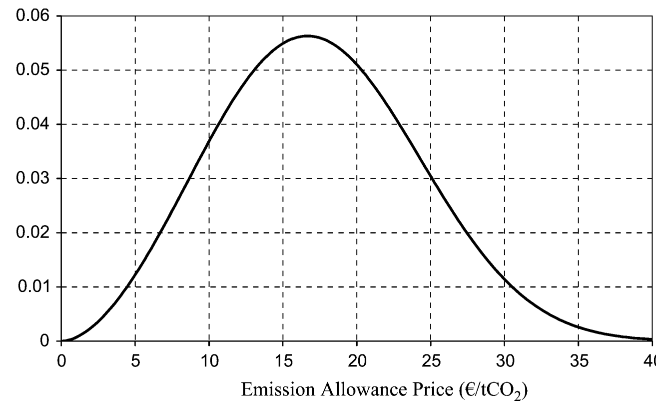

Fig. 3. Weibull PDF modeling CO_2 emission price.

Rather than representing the CO_2 emission price as a single value, a probability density function (PDF) is considered. In this paper, based on historic data presented in [3], the CO_2 emission price is approximated by a Weibull PDF which can be expressed by (31) and is shown in Fig. 3. It should be stressed that raw data have not been available in the course of this work so (31) is just an approximate model for CO_2 emission price:

$$f(x) = \frac{2.78}{19.56} \left(\frac{x}{19.56} \right)^{1.78} e^{-(x/19.56)^{2.78}} . \tag{31}$$

IV. Solution Methodology

A proposal for future network expansion, on the one hand should be able to mitigate the majority of CO_2 emission changes resulting in optimal generation pattern shifts, and on the other hand a minimum transmission investment should be made. In order to achieve these targets assuming a volatile CO_2 emission price, a probabilistic approach using Monte Carlo simulation is applied. Monte Carlo is a simulation method based on repetition which is widely used for probabilistic studies in power systems [1], [20] .

In order to incorporate the volatility of CO_2 emission price into transmission planning, using the Monte Carlo simulation philosophy, the TNP problem is solved for CO_2 emission prices generated randomly based on the PDF (31), this process is repeated many times so that all possible cases are fairly simulated. Dash Xpress is used to solve the mixed-integer optimization problem described in III.

$$P_i^{j(c)} - \frac{\left(\theta_k^{j(c)} - \theta_l^{j(c)}\right)}{X_i} = 0 \quad i = 1, \ldots, N_l \tag{25}$$

$$P_i^{j(c)} - \frac{\left(\theta_k^{j(c)} - \theta_l^{j(c)}\right)}{X_i} \leq M_k . (1 - X_i) \quad i = 1, \ldots, \overline{N_l} \tag{26}$$

$$P_i^{j(c)} - \frac{\left(\theta_k^{j(c)} - \theta_l^{j(c)}\right)}{X_i} \geq M_k . (X_i - 1) \quad i = 1, \ldots, \overline{N_l} \tag{27}$$

TABLE I
MONTE CARLO SIMULATION TNP RESULTS FOR MODEL I AND II

Emission Model	Line #	No Need	100 MW	200 MW	300 MW	400 MW	Most probable Capacity (MW)
Model I	12	0	2000	0	0	0	100
	13	0	2000	0	0	0	100
	18	0	0	0	1070	930	300
	20	0	0	0	1096	904	300
	21	0	0	35	1965	0	300
	22	1722	0	37	241	0	No Need
	31	644	165	270	921	0	300
	34	0	1195	805	0	0	100
	35	0	1195	805	0	0	100
	36	0	0	1196	804	0	200
	37	0	0	1196	804	0	200
	38	647	155	270	928	0	300
Model II	12	0	2000	0	0	0	100
	13	0	2000	0	0	0	100
	18	0	0	0	1759	241	300
	20	0	0	0	2000	0	300
	21	0	0	1153	109	738	200
	22	738	0	1218	44	0	200
	31	137	115	1748	0	0	200
	34	387	1148	464	0	0	100
	35	387	1148	464	0	0	100
	36	0	0	1854	146	0	200
	37	0	0	1854	146	0	200
	38	137	115	1748	0	0	200

The optimum transmission capacity among all resultant capacities of simulated cases is one which has appeared most frequently. In other words, the most frequent optimum capacity can be the best capacity leading to most probable minimum operating cost along with minimum transmission investment.

V. NUMERICAL STUDIES

In this section, a probabilistic TNP study as well as validation of the loss model are performed on the IEEE 24-bus reliability test system [21]. Transmission line data, load levels information, generation types, prices, and emission factors assumed for this system are given in the Appendix.

An annuitized transmission investment cost of 11.79 €/km.MW.year is used. This transmission expansion constant was calculated based on National Grid Report [22] and the currency rate of $£1.00 = €1.23$.

A. Probabilistic TNP Study

In this section the most likely optimum transmission capacities are calculated using Monte Carlo simulation. The mixed-integer problem described in Section III is solved for 2000 different CO_2 emission prices which are generated randomly based on the PDF (31). In this study, 12 candidate lines shaded in Table VII are assumed. Furthermore the credible capacities of each candidate line can be 100, 200, 300 or 400 MW. The study is undertaken for both CO_2 emission models expressed by (29) and (30) referred to as Model I and Model II, respectively, from this point onwards.

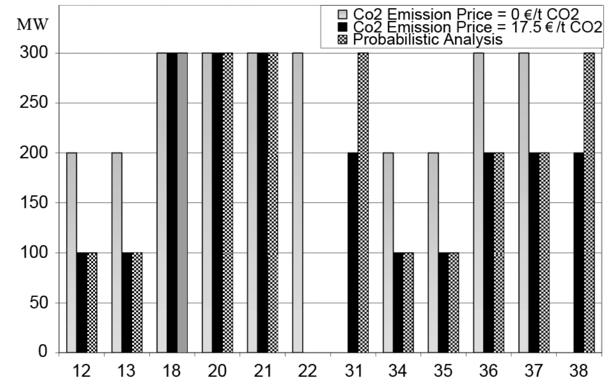

Fig. 4. Results comparison between deterministic studies and probabilistic study considering Model I.

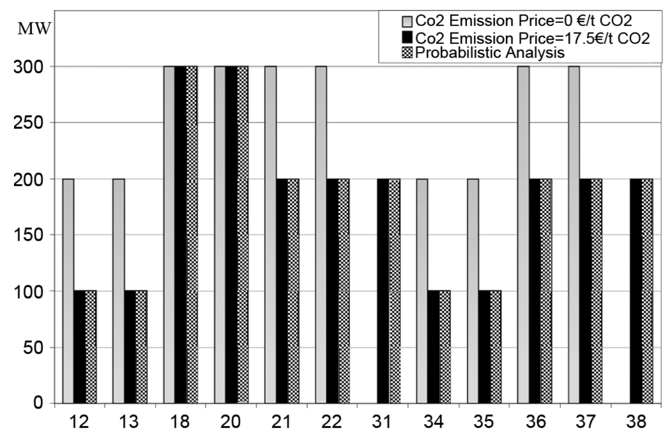

Fig. 5. Results comparison between deterministic studies and probabilistic study considering Model II.

Table I shows the results of Monte Carlo simulation for both emission models. In this table, each row shows the number of times out of 2000 that a particular capacity is allocated to a specific line. For instance, as results show in Table I, considering Model I, for line 18 a capacity of 300 MW is selected 1070 times representing the optimal capacity for 1070 different emission prices. The simulation also demonstrates that 930 times a line with a capacity of 400 MW is required. Therefore, it is more likely that 300 MW is the optimal capacity for line 18. Taking the same approach, the most probable optimal capacities are calculated for all candidate lines, as given in the last column of Table I.

In order to make a comparison and examine the benefit accruing from taking the proposed probabilistic approach, two deterministic TNP studies are conducted. Firstly, CO_2 emission cost is not considered in the course of TNP and secondly the most likely CO_2 emission allowance price, which from (31) is 17.5 $€/tCO_2$ is assumed in TNP.

The aforementioned deterministic studies are undertaken for both Model I and II and the resultant transmission capacities along with results of probabilistic study (Table I) are illustrated in Figs. 4 and 5. Accordingly the investment costs for all case studies are tabulated in Table II. As seen in Fig. 5, transmission capacities proposed by the deterministic study assuming

TABLE II
TRANSMISSION INVESTMENT COSTS (€/YEAR)
CALCULATED FROM DIFFERENT STUDIES

	Probabilistic Study	CO_2 emission price =0 €/t CO_2	CO_2 emission price =17.5 €/t CO2
Model I	1131840	1121229	990360
Model II	1052847	1121229	1052847

a 17.5 $€/tCO_2$ for CO_2 emission price are identical to probability study results. In other studies, however, different transmission capacities have been proposed.

For each set of transmission capacities resulting from deterministic and probabilistic studies different total system costs may materialized. The difference between these total system costs can be used as a comparison criterion to prove that the probabilistic study leads to a better transmission expansion proposal.

Total system cost comprises operating cost as well as transmission investment cost which is given in Table II. The operating cost is also made of generation cost plus CO_2 emission cost. The operating cost, however, for a given network and given CO_2 emission price, can be calculated by solving an optimal power flow (OPF) problem; which is a simplified form of (1). In this formulation the transmission investment term is eliminated and all transmission capacities are known, but the security constraint should still be respected. It is obvious that in this case all decision variables are linear continuous variables.

The total system cost for every transmission network proposed by either probabilistic or deterministic approaches shown in Figs. 4 and 5 can be calculated as follows.

Step 1) Consider a particular network whose transmission capacities are given.
Step 2) Consider a CO_2 emission price.
Step 3) Run OPF.
Step 4) Calculate operating cost.
Step 5) Calculate total system cost which is a summation of resultant operating cost in Step 4 and relevant transmission cost in Table II.

The above process is repeated for all those 2000 random CO_2 emission prices. Therefore 2000 total system costs for each transmission network proposed in Table II can be calculated. In order to compare the probabilistic approach with deterministic methods, normal PDFs can be fitted to differentials between total system costs of network proposed by probabilistic method and network calculated by deterministic method, as shown in Figs. 6–8. If a PDF has a negative mean value it shows that the total system cost of network proposed by probabilistic study is more likely to be lower than the total system cost of network calculated by deterministic approaches.

For model I, as shown in Fig. 6, with a possibility of 77.01% is economically beneficial to take a probabilistic approach to TNP compared to a deterministic approach with considering CO_2 emission price is zero. The probabilistic method still brings more benefit with a possibility of 57.09% even if deterministic approach is used assuming the most probable CO_2 emission price which is 17.5 $€/tCO_2$ (see Fig. 7).

For model II, as illustrated in Fig. 8, the likelihood of accruing benefit is 93.06% if probabilistic method is deployed rather than deterministic approach with eliminated CO_2 emission price. It should be stressed, however, that identical networks proposed by both probabilistic method and deterministic method with a CO_2 emission prices of 17.5 €/t CO_2, so there is no cost difference between these cases.

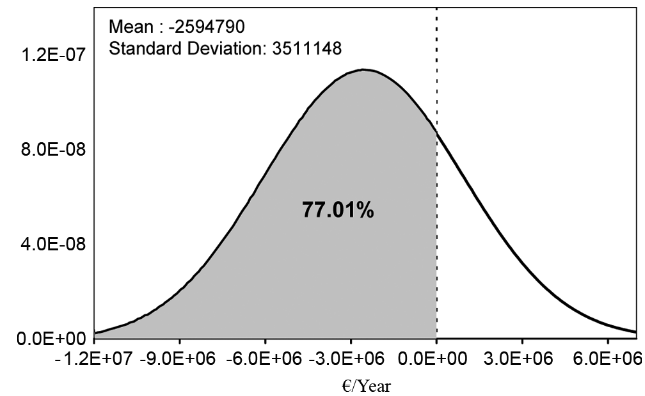

Fig. 6. PDF of total system cost differences between the network proposed by probabilistic analysis and the network proposed by deterministic analysis when emission cost is not considered, Model I.

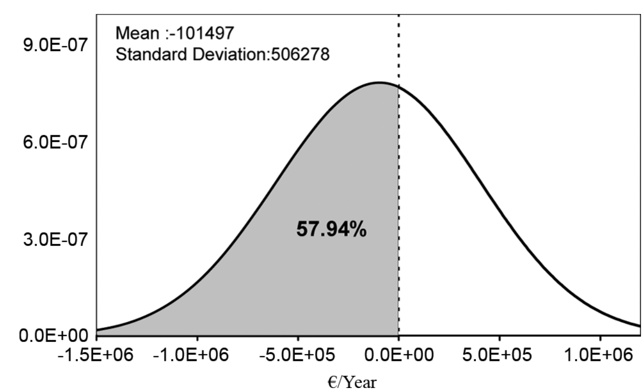

Fig. 7. PDF of total system cost differences between the network proposed by probabilistic analysis and the network proposed by deterministic analysis when emission cost is 17.5 $€/tCO_2$, Model I.

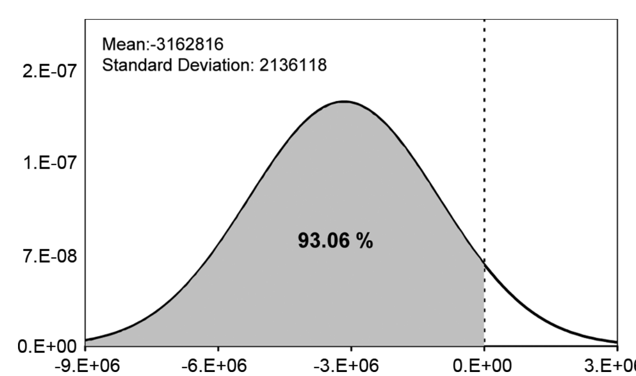

Fig. 8. PDF of total system cost differences between the network proposed by probabilistic analysis and the network proposed by deterministic analysis when emission cost is not considered, Model II.

B. Loss Model Validation

According to the method described in Section III-A, the losses are modeled in the IEEE 24-bus network with two load levels. The model of loss is assumed to have three line segments

TABLE III
CALCULATED LOSSES (MW) IN CASE 1 AND CASE 2

		Case 1	Case 2	Accuracy (%)
Load Levels	1	35.7	34.8	97
	2	31.5	30.7	97

$(N = 3)$. As in this section the main focus is on losses, the effect of CO_2 emission allowance price is eliminated by setting $\pi_{CO_2} = 0$ and the security constraint is also ignored. The maximum capacity of transmission lines is the same as the resultant transmission capacities of the probabilistic study for Model I, given in Table I. By running OPF a comparison is then made between the two different cases.

Case 1) $Ploss_i^j$ is evaluated by deploying the proposed piecewise linear model introduced in Section III-A.

Case 2) Regarding resultant power flow and relevant resistance, $Ploss_i^j$ is calculated by using a quadratic (7).

For both cases the total losses at each load level can be calculated from (32):

$$Loss_{total}^j = \sum_{i=1}^{Nl} Ploss_i^j. \tag{32}$$

Table III presents the comparison between aforementioned cases. As can be seen the resultant losses of the linear model are very close to the losses calculated by nonlinear quadratic function (7). Therefore the proposed piecewise linear loss model can be used with confidence.

VI. CONCLUSION

This paper has addressed the effect of variations in CO_2 emission allowances price on transmission network planning.

The transmission network planning problem was formulated as a mixed-integer optimization problem so as to model the lumpy nature of transmission investment.

As losses are important in transmission network planning, an accurate linear loss model has been proposed and validated in this paper. The losses model has been implemented in the transmission network planning problem which, in this paper, has been formulated as a two-period security constrained dc OPF problem.

Two different emission allowance and trading models have been proposed. In Model I generators can buy or sell their emission allowance in an emissions market and hence both penalties and benefits can accrue to generation units. Model II, however, represents a system under which a generator is penalized if it emits more than its emission allowance cap, however no benefit accrues to units emitting less than the cap. Although it is more likely that Model I represents the real installations behavior in the emission market, but due to some uncertainties in future policies Model II was also investigated.

CO_2 emission price was considered in the form of a probability density function. Hence rather than a deterministic approach a probabilistic approach was taken to transmission network planning. The Monte Carlo method was used to simulate the volatility of CO_2 emission price enabling the most probable optimum transmission capacity to be calculated.

Probabilistic as well as deterministic studies were carried out on the IEEE 24-bus Reliability Test System for both CO_2 emission models. A comparative analysis of the results from these studies has demonstrated that transmission planning based on the probabilistic method is more likely to lead to lower total system costs than transmission planning under the deterministic method.

TABLE IV
GENERATION INFORMATION FOR THE 24-BUS NETWORK

Gen. #	Bus #	Gen. Type	Pmax (MW)	Pmin (MW)	Allowance cap (tCo_2/year)
1	1	Hydro	40	0	--
2	1	Coal	152	0	4800
3	2	Hydro	40	0	--
4	2	Coal	152	0	4800
5	7	CCGT	150	0	4000
6	13	CCGT	591	0	7900
7	15	Hydro	60	0	--
8	15	Coal	155	0	4900
9	16	Coal	155	0	4900
10	18	Nuclear	400	0	--
11	21	Nuclear	400	0	--
12	22	CCGT	300	0	4000
13	23	Coal	310	0	9800
14	23	Coal	350	0	11000

TABLE V
LOAD INFORMATION FOR THE 24 -BUS NETWORK

Bus #	Load Level 1 (MW)	Load Level 2 (MW)	Bus #	Load Level 1 (MW)	Load Level 2 (MW)
1	108	59.4	13	265	145.7
2	97	53.3	14	194	106.7
3	180	99	15	317	174.3
4	74	40.7	16	100	51
5	71	39.0	17	0	0
6	136	74.8	18	333	183.1
7	125	68.7	19	181	99.5
8	171	94	20	128	70.4
9	175	96.2	21	0	0
10	195	107.2	22	0	0
11	0	0	23	0	0
12	0	0	24	0	0
Load Level 1 Duration : 2920 hours					
Load Level 2 Duration : 5840 hours					

APPENDIX

Table IV shows the maximum and minimum generation output limits and CO_2 emission allowances for the 24 bus network. Load level information is given in Table V.

Four different types of generators are assumed in the network. Generation prices from [23] and emission factors from [24] are given in Table VI. Network branch data as well as candidate lines are brought in Table VII.

TABLE VI
GENERATION PRICES AND CO_2 EMISSION FACTORS

Gen. Type	€/MWh	tCo_2/MWh
Coal	45.2	0.88
CCGT	51.2	0.37
Nuclear	35	0
Hydro	50	0

TABLE VII
TRANSMISSION LINE DATA FOR 24-BUS NETWORK

Line #	Sending Bus	Receiving Bus	R(p.u.)*	X(p.u.)*	Capacity (MW)	Length (Km)
1	1	2	0.0026	0.0139	100	3
2	1	3	0.0546	0.2112	100	55
3	1	5	0.0218	0.0845	100	22
4	2	4	0.0328	0.1267	100	33
5	2	6	0.0497	0.192	150	50
6	3	9	0.0308	0.119	150	31
7	3	24	0.0023	0.0839	300	50
8	4	9	0.0268	0.1037	100	27
9	5	10	0.0228	0.0883	100	23
10	6	10	0.0139	0.0605	150	16
11	7	8	0.0159	0.0614	150	16
12	8	9	0.0427	0.1651	---	43
13	8	10	0.0427	0.1651	---	43
14	9	11	0.0023	0.0839	200	50
15	9	12	0.0023	0.0839	300	50
16	10	11	0.0023	0.0839	300	50
17	10	12	0.0023	0.0839	300	50
18	11	13	0.0061	0.0476	---	33
19	11	14	0.0054	0.0418	400	29
20	12	13	0.0061	0.0476	---	33
21	12	23	0.0124	0.0966	---	67
22	13	23	0.0111	0.0865	---	60
23	14	16	0.005	0.0389	400	27
24	15	16	0.0022	0.0173	400	12
25	15	21	0.0063	0.049	400	34
26	15	21	0.0063	0.049	400	34
27	15	24	0.0067	0.0519	300	36
28	16	17	0.0033	0.0259	400	18
29	16	19	0.003	0.0231	300	16
30	17	18	0.0018	0.0144	400	10
31	17	22	0.0135	0.1053	---	73
32	18	21	0.0033	0.0259	150	18
33	18	21	0.0033	0.0259	150	18
34	19	20	0.0051	0.0396	---	27.5
35	19	20	0.0051	0.0396	---	27.5
36	20	23	0.0028	0.0216	---	15
37	20	23	0.0028	0.0216	---	15
38	21	22	0.0087	0.0678	---	47

* The base power is 100 MW

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